

PUPIL NAME: _____ CLASS: _____ TEACHER: _____

KEY TOPIC 2 • C1500–C1700

The Medical Renaissance in England

Humanism, The Anatomical Revolution, and The Persistence of Ancient Dogma



Primary Visual Source: *Andreas Vesalius Performing Human Dissection at Padua* (Frontispiece of *De Fabrica*, 1543)

Accession Shelfmark: **ARCH-MED-1543**

COURSE SPECIFICATION • KEY ENQUIRY SEQUENCE	LEARNT	REVISED
Key Topic 2.1: The New Spirit of Enquiry: Humanism, The Printing Press & The Royal Society <i>How did movable type printing and the scientific method challenge 1,500 years of clerical medical dogma?</i>	☐	☐
Key Topic 2.2: Thomas Sydenham & The Art of Bedside Observation (1676) <i>Why was Sydenham nicknamed 'The English Hippocrates', and how did his disease classification revolutionize diagnosis?</i>	☐	☐
Key Topic 2.3: Andreas Vesalius & The Anatomical Revolution (1543) <i>How did direct human dissection in <i>De Fabrica</i> disprove 300+ Galenic errors, and why did it fail to cure disease?</i>	☐	☐
Key Topic 2.4: William Harvey & The Circulation of the Blood (1628) <i>How did mathematical calculation and ligature experiments prove that blood circulates rather than being burned as fuel?</i>	☐	☐
Key Topic 2.5: Case Study: Continuity in Treatment & The Great Plague of London (1665) <i>How did civic quarantine orders change municipal response, and why did treatment show complete continuity with 1348?</i>	☐	☐

LIVING TIMELINE • PART 1: HUMANISM & THE ANATOMICAL REVOLUTION (1440–1660)

Instructions: As you study each enquiry lesson, complete the timeline missions by sketching and annotating in the corresponding Key Topic boxes below.

1440 • Johannes Gutenberg: Movable Metal Type Printing Press

Key Topic 2.1

Gutenberg invents movable metal type in Mainz (introduced to England by Caxton in 1476). Shatters Church monopoly on medical manuscript copying. Enables rapid mass-production of identical medical texts and anatomical diagrams without copyist errors.

1543 • Andreas Vesalius: De Humani Corporis Fabrica & Padua Dissections

Key Topic 2.3

Vesalius personally conducts human dissections at Padua, correcting over 300 Galenic anatomical errors (e.g. human lower jaw is one bone not two). Masterfully illustrated by Titian's workshop and mass-printed, creating the foundation of modern anatomy.

1628 • William Harvey: De Motu Cordis & Circulation of the Blood

Key Topic 2.4

Harvey mathematically calculates the volume of blood pumped per hour (540 pints), proving blood circulates continuously through arteries and veins. Disproves Galen's 1,400-year theory that blood is consumed as fuel manufactured by the liver.

1660 • The Royal Society: Nullius in Verba & Scientific Peer Review

Key Topic 2.1

Founded under royal patronage (Charles II). Operating under the motto Nullius in Verba ('Take nobody's word for it'), it establishes the empirical scientific method, publishing Philosophical Transactions (1665) to share peer-reviewed laboratory research across Europe.

LIVING TIMELINE • PART 2: OBSERVATION, PLAGUES & THE MICROSCOPIC WORLD (1665–1676)

Instructions: Complete the timeline sketches and notes as you master each enquiry lesson.

1665 • Robert Hooke: Micrographia & Discovery of Plant Cells

Key Topic 2.1

Hooke uses compound microscopes to examine cork tissue, coining the term 'cell'. His stunning fold-out engravings of fleas and insects shock the public, demonstrating that visible nature contains an unseen microscopic architecture.

1665 • The Great Plague of London: Civic Quarantine & Bills of Mortality

Key Topic 2.5

The plague kills 100,000 Londoners. The Mayor enforces strict municipal orders: infected houses padlocked for 28 days with red crosses; parish watchmen guard doors; mass plague pits dug; yet treatments (bleeding, tobacco) remain identical to 1348.

1676 • Thomas Sydenham: Observationes Medicae & Bedside Observation

Key Topic 2.2

The 'English Hippocrates' urges doctors to abandon theoretical books and study living symptoms at the bedside. Classifies diseases into distinct biological species (distinguishing scarlet fever from measles); prescribes cool regimes for smallpox and cinchona bark for malaria.

1676 • Antonie van Leeuwenhoek: Discovery of 'Animalcules' (Bacteria)

Key Topic 2.1

Using precision single-lens microscopes magnifying up to 300x, Leeuwenhoek discovers living microorganisms in rainwater and dental plaque. Yet without understanding that microbes cause illness, the discovery saves zero lives until Pasteur in 1861.

• 'DO NOW' RETRIEVAL DRILL (10 RECALL QUESTIONS)

Score: [/ 10]

1. In what century did Hippocrates live and teach in Ancient Greece?

2. Name the four bodily fluids identified in Hippocratic medicine.

3. Which ancient Roman physician expanded this into the Theory of Opposites?

4. Why did the medieval Catholic Church actively protect Galen's medical ideas?

5. What Latin term refers to foul-smelling air blamed for causing epidemic disease?

6. In what year did the Black Death first arrive in England?

7. Which Franciscan friar was imprisoned in 1277 for urging scientific experiments?

8. What glass flask was used by medieval physicians to inspect urine?

9. What was the primary guiding principle of medieval monastic hospitals?

10. Name one humoral treatment used by medieval physicians to purge excessive blood.

• KEY VOCABULARY TASK

TERMINOLOGY

Define **The Royal Society (1660)** and explain how its motto **Nullius in Verba** challenged medieval clerical authority:

• QUESTION 3(A): EXPLAIN ONE SIMILARITY [4 MARKS]

[4 MARKS • 5 MINS]

Explain one way in which ideas about the cause of disease in the Renaissance (c1500–c1700) were similar to ideas in the Medieval period (c1250–c1500). [4 marks]

Hint: Focus on the persistent reliance on miasma (foul air) and God's will among ordinary people, despite elite scientific debates.

Stems: One way ideas about causes were similar was the continued belief in... • In the Medieval period... • Similarly, in the Renaissance...

• QUESTION 3(B): EXPLAIN ONE DIFFERENCE [4 MARKS]

[4 MARKS • 5 MINS]

Explain one way in which the communication of medical ideas in the Renaissance was different from the Medieval period. [4 marks]

Hint: Contrast the mass-production of identical medical texts on printing presses with hand-copied manuscripts controlled by monks in scriptoria.

Stems: One way communication differed was... • In the Medieval period, books were... • In contrast, in the Renaissance, the printing press...

EXTENDED WRITING ASSESSMENT

Stimulus: The printing press (c1440/1476) The Royal Society (1660/1662) *(You must also use information of your own)*

1. BREAKING CLERICAL MONOPOLY

2. THE ROYAL SOCIETY & PEER REVIEW

3. LIMITS OF CHANGE (CRITERIA)

ANALYTICAL CONNECTIVES:

KEY VOCABULARY BANK:

Task: Using the structure strip above, write 1-2 detailed analytical paragraphs below (continue in your exercise book for full timed paper):

← Pages 2-3

• 'DO NOW' RETRIEVAL DRILL (10 RECALL QUESTIONS)

Score: [/ 10]

1. In what year did Johannes Gutenberg invent the movable metal type printing press?

2. In what year did William Caxton introduce the first printing press to England?

3. How did the printing press break the Catholic Church's clerical monopoly on books?

4. In what year was the Royal Society formally founded in London?

5. What was the Latin motto of the Royal Society, meaning 'Take nobody's word for it'?

6. Name the world's first peer-reviewed scientific journal, published in 1665.

7. Who published *Micrographia* in 1665 featuring detailed microscopic illustrations?

8. What term did Antonie van Leeuwenhoek give to microscopic bacteria in 1676?

9. Why did optical discoveries of microbes save zero lives in the Renaissance?

10. Which ancient Greek physician originated the clinical observation of patients?

• KEY VOCABULARY TASK

TERMINOLOGY

Define **Bedside Observation** and explain how Sydenham's concept of **Disease Species** differed from humoral theory:

• QUESTION 3(A): EXPLAIN ONE DIFFERENCE [4 MARKS]

[4 MARKS • 5 MINS]

Explain one way in which Thomas Sydenham's approach to diagnosis was different from medieval physicians. [4 marks]

Hint: Contrast Sydenham's detailed bedside observation of clinical symptoms with medieval reliance on urine flasks (matula) and astrological charts.

Stems: One way diagnosis differed was... • In the Medieval period, doctors relied on... • In contrast, Sydenham insisted on...

• QUESTION 3(B): EXPLAIN ONE SIMILARITY [4 MARKS]

[4 MARKS • 5 MINS]

Explain one way in which medical treatments prescribed by Thomas Sydenham were similar to medieval treatments. [4 marks]

Hint: Focus on his continued reliance on traditional humoral purging and bloodletting (phlebotomy) to deplete patient fluids.

Stems: One way treatments were similar was the continued use of... • In the Medieval period... • Similarly, Sydenham continued to prescribe...

EXTENDED WRITING ASSESSMENT

Stimulus: **Observationes Medicae (1676)** **Classifying diseases into species** *(You must also use information of your own)*

[illegible]

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• 'DO NOW' RETRIEVAL DRILL (10 RECALL QUESTIONS)

Score: [/ 10]

1. What nickname was given to Thomas Sydenham because of his clinical approach?

2. In what landmark 1676 book did Sydenham set out his clinical observations?

3. How did Sydenham view diseases differently from humoral theory?

4. Which two childhood illnesses did Sydenham successfully prove were separate diseases?

5. What imported South American tree bark did Sydenham prescribe for malaria fevers?

6. What revolutionary cooling treatment did Sydenham prescribe for smallpox patients?

7. In what way did Sydenham's treatments still show continuity with the Middle Ages?

8. What was the Royal Society's Latin motto promoting empirical evidence?

9. Where were medical manuscripts copied prior to the invention of the printing press?

10. In what year did William Caxton set up the first printing press in England?

• KEY VOCABULARY TASK

TERMINOLOGY

Define **Human Dissection** and explain why Vesalius's corrections of **Galenic Anatomy** met furious resistance:

• QUESTION 3(A): EXPLAIN ONE DIFFERENCE [4 MARKS]

[4 MARKS • 5 MINS]

Explain one way in which methods of investigating the human body in the Renaissance were different from methods in the Medieval period. [4 marks]

Hint: Contrast medieval professors lecturing from Galenic texts while barbers cut with Vesalius performing human dissections himself at Padua.

Stems: One way methods of investigation differed was... • In the Medieval period, professors... • In contrast, in the Renaissance, Vesalius...

• QUESTION 3(B): EXPLAIN ONE SIMILARITY [4 MARKS]

[4 MARKS • 5 MINS]

Explain one way in which the effectiveness of surgical treatments in the 16th century was similar to the Medieval period. [4 marks]

Hint: Focus on the fact that despite accurate anatomical diagrams, surgeons still lacked anesthetics, antiseptics, and effective treatments, leaving surgery agonizing and fatal.

Stems: One way the effectiveness of surgery was similar was... • In the Medieval period... • Similarly, in the 16th century...

Stimulus: De Humani Corporis Fabrica (1543) Medical treatments in the 16th century (You must also use information of your own)

1. OVERTURNING GALEN (CHANGE) Explain how Vesalius dissected human cadavers himself at Padua, correcting 300+ errors (e.g. human lower jaw is 1 bone not 2; no invisible heart pores).	2. VISUAL MASS COMMUNICATION Explain the impact of De Fabrica (1543): masterfully illustrated anatomical plates printed identically on printing presses, inspiring a generation of anatomists.	3. LIMITATIONS ON TREATMENT (CRITERIA) Evaluate: Vesalius created accurate anatomy, but knowing the structure of the body did not cure disease. Doctors remained helpless at the bedside.
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Andreas Vesalius • University of Padua • De Humani Corporis Fabrica (1543)
• human dissection • 300+ Galenic errors • mandible (jawbone) • sternum (3 parts not 7) • septum pores • anatomical plates • lack of medical cures • continued humoral bleeding

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

9/14

KT2.4: William Harvey & The Circulation of the Blood (1628)

KNOWLEDGE RETRIEVAL • KEY VOCABULARY • EXAM PRACTICE

• 'DO NOW' RETRIEVAL DRILL (10 RECALL QUESTIONS)

Score: [/ 10]

1. In what year did Andreas Vesalius publish *De Humani Corporis Fabrica*?

2. At which famous Italian university did Vesalius serve as professor of surgery?

3. How did Vesalius's method of teaching anatomy differ from medieval professors?

4. How many anatomical errors made by Galen did Vesalius identify and correct?

5. State one specific anatomical error of Galen corrected by Vesalius.

6. Why did Vesalius's accurate anatomical discoveries fail to cure sick patients?

7. Which ancient Roman doctor's anatomical monopoly was overturned by Vesalius?

8. What 1676 book did Thomas Sydenham publish on clinical bedside symptoms?

9. In what year was the Royal Society granted its Royal Charter by Charles II?

10. What was the Royal Society's motto promoting empirical evidence?

• KEY VOCABULARY TASK

TERMINOLOGY

Define **Circulation of the Blood** and explain how Harvey's **Mechanical Pump** model overturned Galen's liver theory:

• QUESTION 3(A): EXPLAIN ONE DIFFERENCE [4 MARKS]

[4 MARKS • 5 MINS]

Explain one way in which William Harvey's understanding of the circulatory system was different from Galen's theories. [4 marks]

Hint: Contrast Galen's idea that blood is constantly manufactured in the liver and burned as fuel with Harvey's proof of a closed, continuous circulation pumped by the heart.

Stems: One way Harvey's ideas differed was... • Galen claimed that blood was... • In contrast, Harvey proved that...

• QUESTION 3(B): EXPLAIN ONE SIMILARITY [4 MARKS]

[4 MARKS • 5 MINS]

Explain one way in which medical treatments after Harvey's discovery were similar to treatments in the Medieval period. [4 marks]

Hint: Focus on the fact that physicians continued to practice bloodletting (phlebotomy) to balance humours because Harvey's discovery offered zero practical cures.

Stems: One way medical treatments remained similar was... • In the Medieval period... • Similarly, after Harvey's discovery...

Explain why William Harvey's discovery of the circulation of the blood was a turning point in medicine. [12 marks]

Stimulus: Calculating blood volume (540 pints/hr) De Motu Cordis (1628) (You must also use information of your own)

STRUCTURE STRIP • 3-PARAGRAPH ANALYTICAL PLAN

1. THE HEART AS A MECHANICAL PUMP

Explain how Harvey proved the heart acts as a pump, circulating blood through a closed one-way system of arteries and veins.

2. MATHEMATICAL CALCULATION

Explain how Harvey calculated that the heart pumped 540 pints an hour (more than body weight), disproving Galen's theory that the liver made blood from food.

3. RESISTANCE & TREATMENT LIMITS

Explain why conservative doctors rejected Harvey ('quack') and why his discovery did not change medical treatments or halt bloodletting.

ANALYTICAL CONNECTIVES:

Harvey's discovery was a turning point because... • By applying mechanical pump theory, he proved... • Crucially, his mathematical calculations proved... • However, the practical impact was limited because... • Consequently...

KEY VOCABULARY BANK:

William Harvey • De Motu Cordis (1628) • circulation of the blood • mechanical pump • arteries • veins • vein valves (Fabricius) • ligature experiment • 540 pints per hour • capillaries • conservative backlash

Task: Using the structure strip above, write 1–2 detailed analytical paragraphs below (continue in your exercise book for full timed paper):

Handwriting practice lines for the analytical paragraphs.

TIMELINE MISSION

Turn to Pages 2–3 (Key Topic 2.4). In the drawing box, sketch the famous arm ligature experiment showing vein valves and annotate: De Motu Cordis (1628)!

← Pages 2–3

• 'DO NOW' RETRIEVAL DRILL (10 RECALL QUESTIONS)

Score: [/ 10]

1. In what year did William Harvey publish De Motu Cordis?

2. How did Harvey mathematically disprove Galen's liver theory?

3. What one-way valves in veins did Harvey investigate using his arm ligature experiment?

4. What microscopic vessels connecting arteries and veins could Harvey not observe?

5. Why did conservative doctors nickname Harvey a 'circulator' (quack)?

6. In what year was Vesalius's De Humani Corporis Fabrica published?

7. State one anatomical error of Galen corrected by Vesalius.

8. What nickname was given to Thomas Sydenham?

9. In what year did the Black Death first strike England?

10. Name the imported South American tree bark prescribed by Sydenham for malaria fevers.

• KEY VOCABULARY TASK

TERMINOLOGY

Define **Civic Quarantine** and explain how the 1665 response to **Miasma** showed continuity with 1348:

• QUESTION 3(A): EXPLAIN ONE SIMILARITY [4 MARKS]

[4 MARKS • 5 MINS]

Explain one way in which attempts to prevent the spread of the Great Plague (1665) were similar to attempts during the Black Death (1348). [4 marks]

Hint: Focus on the continued belief in miasma, leading people to carry pomanders, smoke tobacco, and light street bonfires, alongside prayers to avert God's wrath.

Stems: One way attempts to prevent spread were similar was... • During the Black Death in 1348... • Similarly, during the Great Plague of 1665...

• QUESTION 3(B): EXPLAIN ONE DIFFERENCE [4 MARKS]

[4 MARKS • 5 MINS]

Explain one way in which municipal and civic reactions to the Great Plague (1665) were different from reactions to the Black Death (1348). [4 marks]

Hint: Contrast the absence of organized municipal quarantine in 1348 with strict 28-day house lock-ins, red crosses on doors, parish watchmen, and Bills of Mortality in 1665.

Stems: One way civic reactions differed was... • During the Black Death in 1348... • In contrast, during the Great Plague of 1665, the Mayor of London...

Stimulus: The Four Humours and miasma Quarantine and watchmen *(You must also use information of your own)*

1. LOCAL CIVIC ACTION (CHANGE) Explain how London authorities enforced organized quarantine: padlocking infected houses for 28 days with red crosses ('Lord have mercy upon us'), watchmen, and Bills of Mortality.	2. MEDICAL TREATMENT (CONTINUITY) Explain how doctors still blamed miasma, God, and the Four Humours: smoking tobacco, holding pomanders, bleeding patients, and taking useless quack 'Plague Water'.	3. EVALUATIVE VERDICT (CRITERIA) Weigh civic organisation against medical ignorance: government responses became more organized, but medical knowledge of the disease was identical to 1348.
ANALYTICAL CONNECTIVES: <i>On the one hand, responses in 1665 showed change because... • For example, local authorities enforced... • On the other hand, there was complete continuity in... • Crucially, medical treatments... • Overall, while civic methods changed...</i>		
KEY VOCABULARY BANK: The Great Plague (1665) • London Bills of Mortality • 28-day house quarantine • red crosses • Lord have mercy upon us • searchers of the dead • mass plague pits • parish watchmen • miasma • pomanders • smoking tobacco • continued bloodletting		

[illegible]

Turn to Pages 2–3 (Key Topic 2.5). In the drawing box, sketch Londoners fleeing the 1665 plague, padlocked houses with red crosses, and the dead-cart collecting bodies. Annotate why treatments showed continuity with 1348! ← **Pages 2–3**

STUDENT ASSESSMENT RECORD & PROGRESS TRACKER

PUPIL NAME: _____ CLASS: _____ TARGET GRADE: _____

Date Completed	Lesson • Specification Focus	Q3 Practice [4m]	Extended Response	Lesson Total
	KT2.1: Printing & Royal Society	Q3 Sim: [/ 4]	Q4 Why: [/ 12]	[/ 16]
	KT2.2: Thomas Sydenham	Q3 Diff: [/ 4]	Q4 Why: [/ 12]	[/ 16]
	KT2.3: Andreas Vesalius	Q3 Diff: [/ 4]	Q5/6 Essay: [/ 20]	[/ 24]
	KT2.4: William Harvey	Q3 Diff: [/ 4]	Q4 Why: [/ 12]	[/ 16]
	KT2.5: Great Plague of 1665	Q3 Sim: [/ 4]	Q5/6 Essay: [/ 20]	[/ 24]
CUMULATIVE ASSESSMENT TOTALS		Q3 Total: [/ 20]	Extended: [/ 76]	[/ 96]

TEACHER FORMATIVE ASSESSMENT • FEEDBACK

Effort: [1 • 2 • 3 • 4 • 5]

WHAT WENT WELL (WWW):

EVEN BETTER IF (EBI):

Teacher Signature: _____Date: _____

INTERACTIVE DIGITAL QUIZZING HUB • SCAN FOR INSTANT RETRIEVAL PRACTICE

ONLINE RECALL

KT2.1
Printing & Science

SCAN TO QUIZ

Best Score: [/ 8]

KT2.2
Thomas Sydenham

SCAN TO QUIZ

Best Score: [/ 8]

KT2.3
Andreas Vesalius

SCAN TO QUIZ

Best Score: [/ 8]

KT2.4
William Harvey

SCAN TO QUIZ

Best Score: [/ 8]

KT2.5
Great Plague 1665

SCAN TO QUIZ

Best Score: [/ 8]